

Grade 6 Arduino — Voice Recognition Module

Lesson Plan & Student Notes

Learn how your Arduino can listen and understand!

LESSON OVERVIEW

Subject	STEM / Computing / Electronics
Grade Level	Grade 6
Duration	60–75 minutes
Topic	Arduino Voice Recognition Module — how computers hear and respond to voice commands
Key Vocabulary	Voice recognition, Arduino, module, digital signal, serial communication, TX/RX pins, LED, void setup(), void loop(), digitalWrite()
Materials	Arduino Uno, voice recognition module, jumper wires, USB cable, optional: LED, speaker, small motor/fan

LEARNING OBJECTIVES

1. Explain what a voice recognition module is and how it works with an Arduino.
2. Describe the steps involved in converting a spoken word into a digital signal the Arduino can process.
3. Identify and name the hardware components required to build a voice-controlled project.
4. Connect a voice recognition module to an Arduino using the correct pins (power, ground, TX, RX).
5. Read and understand a basic Arduino sketch that responds to voice commands.
6. Design and propose an original voice-controlled project using Arduino.
7. Apply safety and best practices when handling electronic components.

What Is a Voice Recognition Module?

A **voice recognition module** is a small electronic device that can listen to spoken words and convert them into signals that a microcontroller (like an Arduino) can understand. Think of it as giving your Arduino a pair of ears — it captures your voice, processes the sound waves, and tells the Arduino exactly what command you gave.

The most common module used with Arduino for beginners is the **V3 Voice Recognition Module**, which can store up to 80 voice commands and recognise up to 7 at a time. It communicates with the Arduino

through a serial connection (TX/RX pins).

Feature	Detail
Max stored commands	Up to 80 voice commands
Active recognition	Up to 7 commands at one time
Communication	Serial (UART) via TX and RX pins
Operating voltage	3.3 V – 5 V (compatible with Arduino Uno)
Recognition accuracy	Speaker-dependent (trained to your voice)

Table 1 – Key specifications of a typical voice recognition module.

How Does Voice Recognition Work?

Voice recognition may seem like magic, but it follows a clear sequence of steps. Here is how your voice travels from your mouth to the Arduino action:

- 1 You speak a command close to the microphone on the voice module.
- 2 The microphone captures the sound wave as an analogue electrical signal.
- 3 The module's built-in processor converts the analogue signal into a digital signal (analogue-to-digital conversion).
- 4 The digital signal is compared against voice patterns stored in the module's memory (trained commands).
- 5 If a match is found, the module sends the corresponding command code to the Arduino via the TX pin (serial communication).
- 6 The Arduino reads the code on its RX pin and executes the programmed action (e.g. turning on an LED, spinning a motor).

The module is **speaker-dependent** — this means you must first *train* it by recording your voice saying each command. Once trained, it will only recognise that particular voice pattern, making it more accurate but less flexible than cloud-based AI assistants.

Components You Need

Component	Purpose	Notes
Voice Recognition Module	Listens to voice; converts to digital signal	V3 module recommended for beginners
Arduino Uno (or similar)	Microcontroller that runs your code	Any 5V Arduino board works

Component	Purpose	Notes
Jumper Wires	Connect components together	Male-to-female wires for module connections
USB Cable	Power the Arduino; upload code	Type-B USB (standard Arduino cable)
LED + 220Ω Resistor	Visual output for testing commands	Any colour LED; resistor protects it
Speaker (optional)	Audio feedback for voice commands	Small 8Ω speaker; needs amplifier
Small Motor/Fan (optional)	Demonstrates physical control	Use a motor driver module (L298N)

Table 2 – Components required for the voice recognition project.

Basic Setup and Connections

Follow these steps carefully to wire the voice recognition module to your Arduino. Always **disconnect power before making or changing connections**.

Module Pin	Arduino Pin	Wire Colour (suggested)	Purpose
VCC	5V	Red	Power supply
GND	GND	Black	Ground / common reference
TX	Pin 2 (RX)	Yellow	Module sends data to Arduino
RX	Pin 3 (TX)	Green	Arduino sends data to module

Table 3 – Pin connections between the voice module and Arduino Uno.

Step-by-step setup checklist:

- 1 Place the Arduino and voice module side-by-side on your workspace.
- 2 Connect VCC on the module to the 5V pin on the Arduino (red wire).
- 3 Connect GND on the module to GND on the Arduino (black wire).
- 4 Connect TX on the module to digital pin 2 on the Arduino (yellow wire).
- 5 Connect RX on the module to digital pin 3 on the Arduino (green wire).
- 6 Connect an LED with a 220Ω resistor between pin 13 and GND for testing.

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Double-check all connections before plugging in the USB cable.

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Connect the Arduino to your computer via USB and open the Arduino IDE.

Simple Coding Example

Here is a basic Arduino sketch that listens for voice commands and controls an LED on pin 13. The module sends a number (1 or 2) over serial when it recognises a trained command.

```
#include <SoftwareSerial.h>

SoftwareSerial voiceSerial(2, 3); // RX=pin2, TX=pin3
int ledPin = 13;

void setup() {
  Serial.begin(9600);           // Serial monitor
  voiceSerial.begin(9600);      // Voice module baud rate
  pinMode(ledPin, OUTPUT);
  Serial.println("Ready! Speak a command.");
}

void loop() {
  if (voiceSerial.available()) {
    int command = voiceSerial.read();
    if (command == 1) {         // Trained as 'LED ON'
      digitalWrite(ledPin, HIGH);
      Serial.println("LED turned ON");
    }
    if (command == 2) {         // Trained as 'LED OFF'
      digitalWrite(ledPin, LOW);
      Serial.println("LED turned OFF");
    }
  }
}
```

Understanding the Code:

#include <SoftwareSerial.h>	Loads the SoftwareSerial library so we can use digital pins 2 and 3 as a serial port.
voiceSerial(2, 3)	Creates a serial connection: pin 2 receives data from the module (RX), pin 3 sends data (TX).
Serial.begin(9600)	Starts communication at 9600 baud — both the monitor and the voice module must use the same speed.
voiceSerial.available()	Checks whether the voice module has sent any data (returns true when a command is detected).
command == 1 / 2	The module sends a number matching the trained command slot. Slot 1 = 'LED ON', slot 2 = 'LED OFF'.
digitalWrite(ledPin, HIGH/LOW)	Turns the LED on (HIGH = 5V) or off (LOW = 0V).

Fun Project Ideas

■ Project 1: Voice-Controlled Light

Say "Light on" or "Light off" to control a room lamp. This starter project teaches you how to connect the voice module to an LED (or relay for a real lamp) and respond to your commands. Perfect for practising the basic wiring and coding skills.

- Connect voice module + LED to Arduino.
- Train commands: slot 1 = 'Light on', slot 2 = 'Light off'.
- Upload the sketch; test using the Serial Monitor.
- Extension: use a relay module to control a real lamp.

■ Project 2: Voice-Activated Fan

Say "Fan on" or "Fan off" to spin a small DC motor connected to a fan blade. You will learn about motors, motor driver modules (L298N), and how voice commands can trigger physical movement.

- Connect voice module + L298N motor driver + DC motor.
- Train commands: slot 1 = 'Fan on', slot 2 = 'Fan off'.
- Use `digitalWrite()` to enable/disable the motor driver.
- Extension: add speed control with `analogWrite()` and PWM.

■ Project 3: Voice-Enabled Robot

Build a small wheeled robot and navigate it with your voice! Say "Go forward", "Go back", "Turn left", or "Turn right" to control your robot. This project combines motors, sensors, and voice recognition into one complete creation.

- Use two DC motors and an L298N driver for movement.
- Train 4 commands: forward / back / left / right.
- Map each command to motor direction in your sketch.
- Extension: add an ultrasonic sensor for obstacle avoidance.

REAL-WORLD APPLICATIONS

Application	Technology Used	How It Works
Smart Home Assistants (Alexa, Google Home)	Cloud-based AI + microphone	Converts voice to text, sends to cloud for intent recognition, returns action.
Voice-Controlled Factory Robots	Industrial microcontrollers	Workers issue hands-free commands to robotic arms for safety and efficiency.
Hands-Free Devices for Disabilities	Speech-to-text engines	People with limited mobility control computers, wheelchairs, and home devices by voice.
Voice Commands in Cars	Embedded speech processors	Drivers control GPS, music, and calls without taking hands off the wheel.
Medical Dictation Systems	NLP + speech recognition	Doctors dictate notes directly into electronic health records, saving time.

Table 4 – Real-world uses of voice recognition technology.

Safety and Best Practices



Always ask an adult or teacher before powering on your circuit for the first time.



Double-check all wire connections before connecting to USB power — an incorrect connection can damage components.



Keep your workspace clean and dry. Liquids near electronics can cause short circuits or electric shocks.



Do not connect more components than the Arduino can power (max 200 mA total from 5V pin). Use an external power supply for motors.



Handle all components gently. Arduino pins and module connectors can bend or break if forced.



Never connect the Arduino's 5V pin directly to a motor — always use a motor driver module.



If a component gets hot unexpectedly, disconnect power immediately and check your wiring.

REVIEW QUESTIONS

Knowledge

1. What does a voice recognition module do?
2. Name three components you need to build a voice-controlled project.
3. Which Arduino pins are used to receive data from the voice module?

Understanding

1. Explain in your own words how a spoken word becomes an action on the Arduino.
2. Why does the voice module need to be 'trained' before it can recognise commands?
3. What is the purpose of the SoftwareSerial library in the coding example?

Application

1. A student wants to control three devices (light, fan, buzzer) with three voice commands. How would you set up the voice module and sketch?
2. If the LED does not respond when you say the command, list three things you would check to debug the problem.

Extension (Creative)

1. Design your own voice-controlled invention. Describe what it does, what components it uses, and write pseudocode (plain English steps) for how the Arduino would respond to two commands.

KEY VOCABULARY GLOSSARY

Term	Definition
Arduino	An open-source microcontroller board used to read inputs (sensors, buttons) and control outputs (LEDs, motors).
Voice Recognition Module	A hardware device that captures spoken words and converts them into digital signals for a microcontroller.
Serial Communication	A method of sending data one bit at a time over a wire — the module and Arduino communicate this way.
TX / RX	Transmit (TX) and Receive (RX) pins used for serial data transfer between devices.
Baud Rate	The speed of serial communication, measured in bits per second (e.g. 9600 baud = 9600 bits/s).
void setup()	An Arduino function that runs once when the board powers on — used to initialise settings.
void loop()	An Arduino function that runs repeatedly, checking for inputs and executing actions continuously.
digitalWrite()	An Arduino command that sets a digital pin HIGH (5V/on) or LOW (0V/off).
SoftwareSerial	An Arduino library that allows any digital pins to act as a serial port (not just the built-in TX/RX pins).
Analogue-to-Digital Conversion (ADC)	The process of converting a continuous analogue signal (sound wave) into discrete digital numbers.
Speaker-Dependent Recognition	A voice system trained on one specific person's voice — it recognises only that person's pronunciation.
Motor Driver (L298N)	An electronic module that allows an Arduino to control the speed and direction of DC motors safely.

LESSON SUMMARY — YOU CAN DO IT!

- ✓ A **voice recognition module** gives your Arduino the ability to hear and respond to spoken commands.
- ✓ Voice is captured as an analogue signal, converted to digital, matched against trained commands, and sent to the Arduino via serial communication.

- ✓ You need: voice module, Arduino Uno, jumper wires, USB cable, and optional outputs (LED, fan, robot).
- ✓ The **SoftwareSerial** library handles communication between the module and Arduino on digital pins 2 and 3.
- ✓ Always **train** the module with your own voice before testing and double-check wiring before powering on.
- ✓ Real-world voice technology powers smart assistants, factory robots, accessibility tools, and in-car controls.
- ✓ Keep exploring — your next invention might be the one that changes the world! ■

"You learned about voice modules, connections, coding, and fun projects. Keep exploring and creating amazing things with technology!"